Worksheet 3D: Graphing Quadratics — Solutions

AIML Foundations Mathematics
Dublin and Dún Laoghaire ETB
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Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A

1. 9, 4, 1, 0, 1, 4, 9
2. (a) (0, 0) (b) Yes, line $x = 0$ (c) Upward (d) 0
3. (a) One solution: $x = 0$ (b) No real solutions
4. -9, -4, -1, 0, -1, -4, -9
5. Reflected over x-axis
6. (a) (0, 0) (b) Downward (c) Maximum value 0
7. upward; downward

Part B

8. f: 4, 1, 0, 1, 4; g: 8, 2, 0, 2, 8
9. Doubles them
10. f: 4, 1, 0, 1, 4; h: 2, 0.5, 0, 0.5, 2
11. Halves them
12. $y = 3x^2$
13. $y = 0.25x^2$
14. B, C, A, D
15. up, down; narrower, wider

Part C

16. f: 4, 1, 0, 1, 4; g: 7, 4, 3, 4, 7; h: 2, -1, -2, -1, 2
17. Shifted up 3 units
18. Shifted down 2 units
19. (a) (0, 0) (b) (0, 5) (c) (0, -4) (d) (0, 100)

20. positive; negative
21. (a) $y = x^2 + 7$ (b) $y = x^2 - 3$
22. (a) 0 (b) 1 (c) 2
23. $x = \pm 2$; these are the x-intercepts

Part D

24. -2, -1, 0, 1, 2 $\rightarrow$ 4, 1, 0, 1, 4
25. (2, 0)
26. -2, -1, 0, 1, 2 $\rightarrow$ 4, 1, 0, 1, 4
27. (-3, 0)
28. right; left
29. (a) (5, 0) (b) (-1, 0) (c) (10, 0) (d) (-7, 0)
30. (a) $y = (x-4)^2$ (b) $y = (x+2)^2$
31. (3, 0); (-4, 0)

Part E

32. (a) 2 (b) 1 (c) 3 (d) (1, 3) (e) up (f) narrower
33. (a) -0.5 (b) -2 (c) -1 (d) (-2, -1) (e) down (f) wider
34. (a) $y = (x-3)^2 + 2$ (b) $y = -(x+1)^2 + 4$ (c) $y = 2x^2 - 5$ (d) $y = -0.5(x-2)^2 - 3$
35. A, B, C, D
36. (a) (2, -4) (b) up (c) 0 (d) 0, 4
37. (a) (-1, 9) (b) down (c) 8 (d) -4, 2
38. $y = -(x-2)^2 + 5$
39. $y = (x+1)^2 - 4$
40. minimum, k, h; maximum, k, h
41. (a) 50 m (b) $t = 3$ s (c) 5 m

Part F

42. (a) 1, -6, 5 (b) 3 (c) -4 (d) (3, -4)
43. (a) (-2, -1) (b) 3 (c) -1, -3
44. (a) (1, 9) (b) maximum (c) -2, 4
45. (a) (-2, -2) (b) -1, -3

46. (a) $y = (x+3)^2 + 1$ (b) $y = (x-1)^2 - 4$
47. (a) $y = (x+4)^2 + 4$ (b) $y = (x-5)^2 - 4$
48. (a) $y = x^2 + 4x - 1$ (b) $y = 3x^2 - 6x + 7$
49. (a) $y = -x^2 + 8x - 6$ (b) $y = 2x^2 + 12x + 11$

Part G

50. (a) 4, two (b) 0, one (c) -4, none
51. (a) 1, 3; (b) 2 (double); (c) no real roots
52. (a) $|b| > 6$ (b) $b = \pm 6$ (c) $|b| < 6$
53. $k = 9$
54. (a) 2 (b) 1 (c) 0
55. never crosses the x-axis; vertex (-1, 4)
56. (Sketching)
57. (a) $y = (x-3)^2 - 4$ (b) Yes, vertex below x-axis (c) 1, 5
58. (a) $y = (x-2)^2 + 5$ (b) No, vertex above x-axis
59. (a) $y = -2(x+1)^2 + 3$ (b) Yes, opens down from above (c) $-1 \pm \sqrt{1.5}$

Part H

60. (a) $h = -5(t-2)^2 + 21$ (b) 21 m (c) $t = 2$ s (e) $t \approx 4.05$ s
61. (a) (6, 18); max profit at 600 units (b) 600 units (c) €18,000 (d) 300 and 900 units
62. (a) 50 m (b) 5000 m^2
63. (a) €20 (b) €10,000 (c) €0 and €40
64. (a) 25 m (b) 25 m
65. (a) 2 m (b) 4 m

— ## GeoGebra Quick Reference

To graph in GeoGebra (geogebra.org/calculator):

66. Type equation directly: `y = x^{}2` or `y = 2(x-3)^{}2 + 1`
67. Use `^{}` for exponents: `x^{}2` means x^2
68. Use `\{}textit{` for multiplication: `2}x^{}2` or just `2x^{}2`
69. Add sliders: Type `a = 1` then use a in equations

Useful explorations:

- Type `y = a*x^{}2` and create slider for a - Type `y = (x - h)^{}2 + k` and create sliders for h and k - Type `y = a*(x - h)^{}2 + k` for full control —